AMPLENESS DETECTS NONZERO OBJECTS OF THE DERIVED CATEGORY BY TWISTED COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- A scheme with an ample invertible module is quasi-compact and separated, and objects of the derived category of quasi-coherent modules are represented by complexes of quasi-coherent modules. invoked at proof lines 4294-4302. Sets up the complex representative F^bullet of K.
- Sections of a quasi-coherent module over the affine open of a section of a power of an ample sheaf lift to global sections of the module twisted by a further power of the sheaf. invoked at proof lines 4314-4321. Lifts the local class to a global twisted class in two steps.

Source reference. Original definition 01PS: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4295-4298 for the quasi-compactness and separatedness; the representation of K is inside the counted span.

Source reference. Original lemma 01PY: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4296-4298 together with the definition; the application of the affine diagonal proposition is inside the counted span.

Source reference. Original proposition 08DB: sources/stacks-perfect/pinned-original-source.tex, source lines 1801–1801. Cited at proof lines 4299-4302 to represent K by F^bullet; the class-spreading argument is inside the counted span.

Source reference. Original lemma 01PW: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4314-4321 twice to lift the local class globally; the construction of the nonzero cohomology class is inside the counted span.

Lemma 0.2. *Let X be a scheme and $\mathcal{L}$ an ample invertible $\mathcal{O}_X$-module. If K is a nonzero object of $D_{QCoh}(\mathcal{O}_X)$, then for some $n \geq 0$ and $p \in \mathbf{Z}$ the cohomology group $H^p(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$ is nonzero.*

Proof. Recall that as X has an ample invertible sheaf, it is quasi-compact and separated (Properties, Definition 01PS and Lemma 01PY). Thus we may apply Proposition 08DB and represent K by a complex $\mathcal{F}^\bullet$ of quasi-coherent modules. Pick any p such that $\mathcal{H}^p = \mathrm{Ker}(\mathcal{F}^p \to \mathcal{F}^{p+1})/\mathrm{Im}(\mathcal{F}^{p-1} \to \mathcal{F}^p)$ is nonzero. Choose a

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

point $x \in X$ such that the stalk $\mathcal{H}^p_x$ is nonzero. Choose an $n \geq 0$ and $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that X_s is an affine open neighbourhood of x. Choose $\tau \in \mathcal{H}^p(X_s)$ which maps to a nonzero element of the stalk $\mathcal{H}^p_x$; this is possible as $\mathcal{H}^p$ is quasi-coherent and X_s is affine. Since taking sections over X_s is an exact functor on quasi-coherent modules, we can find a section $\tau' \in \mathcal{F}^p(X_s)$ mapping to zero in $\mathcal{F}^{p+1}(X_s)$ and mapping to τ in $\mathcal{H}^p(X_s)$. By Properties, Lemma 01PW there exists an m such that $\tau' \otimes s^{\otimes m}$ is the image of a section $\tau'' \in \Gamma(X, \mathcal{F}^p \otimes \mathcal{L}^{\otimes mn})$. Applying the same lemma once more, we find $l \geq 0$ such that $\tau'' \otimes s^{\otimes l}$ maps to zero in $\mathcal{F}^{p+1} \otimes \mathcal{L}^{\otimes (m+l)n}$. Then τ'' gives a nonzero class in $H^p(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{L}^{(m+l)n})$ as desired. $\square$